Hands-On Practice Assignment

Practical Lab Tasks

Task 1: Filesystem Structure & File Management

1. Identify your current working directory using pwd.
2. Create a new directory structure named assignment_lab/stage1 using a single command (mkdir -p).
3. Navigate into assignment_lab/stage1 and create three blank text files: data1.txt, data2.txt, and info.txt.
4. Write the names of 5 states into data1.txt using cat > data1.txt or echo.
5. Create a duplicate backup of data1.txt named data1_backup.txt.
6. Create a soft link on your home directory named state_link pointing to data1.txt.

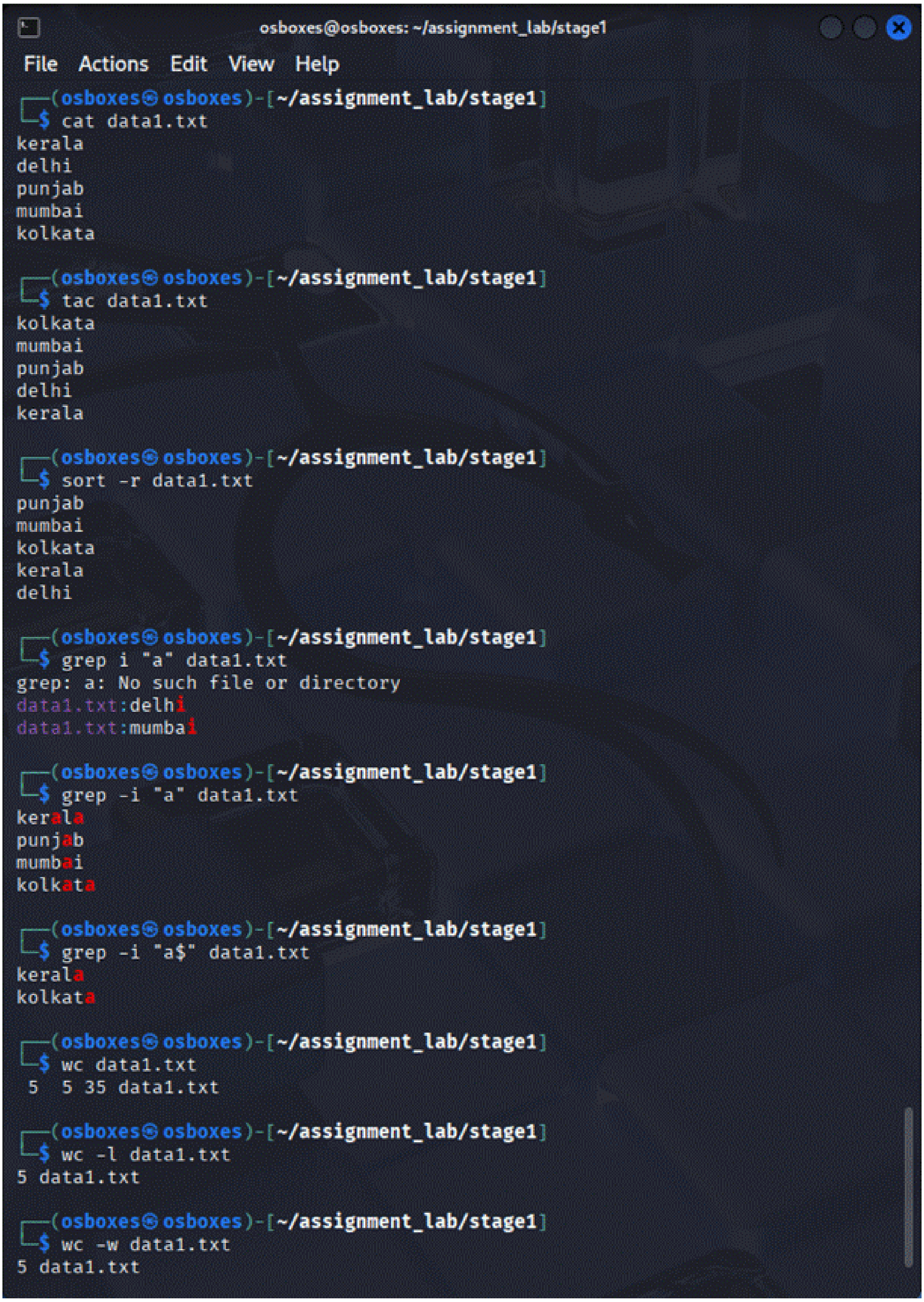
osboxes@osboxes: ~/assignment_lab/stage1

File Actions Edit View Help

```
┌──(osboxes㉿osboxes)-[~]
└─$ pwd
/home/osboxes

┌──(osboxes㉿osboxes)-[~]
└─$ mkdir -p assignment_lab/stage1

┌──(osboxes㉿osboxes)-[~]
└─$ cd assignment_lab/stage1

┌──(osboxes㉿osboxes)-[~/assignment_lab/stage1]
└─$ touch data1.txt data2.txt info.txt

┌──(osboxes㉿osboxes)-[~/assignment_lab/stage1]
└─$ ls -l
total 0
-rw-rw-r-- 1 osboxes osboxes 0 Aug 24 05:47 data1.txt
-rw-rw-r-- 1 osboxes osboxes 0 Aug 24 05:47 data2.txt
-rw-rw-r-- 1 osboxes osboxes 0 Aug 24 05:47 info.txt

┌──(osboxes㉿osboxes)-[~/assignment_lab/stage1]
└─$ cat> data1.txt << 'EOF'
> kerala
> delhi
> punjab
> mumbai
> kolkata
> EOF

┌──(osboxes㉿osboxes)-[~/assignment_lab/stage1]
└─$ cp data1.txt data1_backup.txt

┌──(osboxes㉿osboxes)-[~/assignment_lab/stage1]
└─$ ls
data1_backup.txt  data1.txt  data2.txt  info.txt

┌──(osboxes㉿osboxes)-[~/assignment_lab/stage1]
└─$ cat data1.txt
kerala
delhi
punjab
mumbai
kolkata

┌──(osboxes㉿osboxes)-[~/assignment_lab/stage1]
└─$ ln -s "$PWD/data1.txt" ~/state_link

┌──(osboxes㉿osboxes)-[~/assignment_lab/stage1]
└─$ ls -l ~/state_link
lrwxrwxrwx 1 osboxes osboxes 45 Aug 24 05:53 /home/osboxes/state_link → /hom
e/osboxes/assignment_lab/stage1/data1.txt

┌──(osboxes㉿osboxes)-[~/assignment_lab/stage1]
```

Task 2: Data Extraction & Sorting

1. Display the contents of data1.txt from top to bottom using cat, and then in reverse order using tac.
2. Sort data1.txt in reverse alphabetical order using sort -r.
3. Search for state names ending with or containing the letter a using grep -i.
4. Count the total words and lines present in data1.txt using wc.



Task 3: Archival, Compression & Verification

1. Bundle the entire assignment_lab folder into an archive file named assignment.tar using tar -cvf.
2. Compress assignment.tar using gzip to generate assignment.tar.gz.
3. Use file to verify the compression type of assignment.tar.gz.
4. Preview the decompressed file structure using zcat or tar -tvf.

```
┌──(osboxes㉿osboxes)-[~/assignment_lab/stage1]
└─$ cd ~

┌──(osboxes㉿osboxes)-[~]
└─$ tar -cvf assignment.tar assignment_lab
assignment_lab/
assignment_lab/stage1/
assignment_lab/stage1/info.txt
assignment_lab/stage1/data2.txt
assignment_lab/stage1/data1.txt
assignment_lab/stage1/data1_backup.txt

┌──(osboxes㉿osboxes)-[~]
└─$ ls -lh assignment.tar
-rw-rw-r-- 1 osboxes osboxes 10K Aug 24 06:02 assignment.tar

┌──(osboxes㉿osboxes)-[~]
└─$ gzip assignment.tar

┌──(osboxes㉿osboxes)-[~]
└─$ ls -lh assignment.tar.gz
-rw-rw-r-- 1 osboxes osboxes 291 Aug 24 06:02 assignment.tar.gz

┌──(osboxes㉿osboxes)-[~]
└─$ tar -tvf assignment.tar.gz
drwxrwxr-x osboxes/osboxes   0 2026-08-24 05:46 assignment_lab/
drwxrwxr-x osboxes/osboxes   0 2026-08-24 05:50 assignment_lab/stage1/
-rw-rw-r-- osboxes/osboxes   0 2026-08-24 05:47 assignment_lab/stage1/info.
t
-rw-rw-r-- osboxes/osboxes   0 2026-08-24 05:47 assignment_lab/stage1/data2
xt
-rw-rw-r-- osboxes/osboxes  35 2026-08-24 05:49 assignment_lab/stage1/data1
xt
-rw-rw-r-- osboxes/osboxes  35 2026-08-24 05:50 assignment_lab/stage1/data1
ackup.txt

┌──(osboxes㉿osboxes)-[~]
└─$ █
```

Task 4: System Diagnosis & Remote Monitoring

1. Check your active username with whoami and system uptime with w.
2. Calculate 1542 \times 8 directly from the command line using bc.
3. Verify network reachability to 127.0.0.1 (localhost) sending exactly 3 packets with ping.

```
┌──(osboxes㉿osboxes)-[~]
└─$ whoami
osboxes

┌──(osboxes㉿osboxes)-[~]
└─$ w
 06:09:48 up  1:07,  1 user,  load average: 0.34, 0.19, 0.11
USER     TTY      FROM             LOGIN@   IDLE   JCPU   PCPU  WHAT
osboxes           -                05:03           0.00s  0.01s lightdm --

┌──(osboxes㉿osboxes)-[~]
└─$ echo "1542 * 8" | bc
12336

┌──(osboxes㉿osboxes)-[~]
└─$ ping -c 3 127.0.0.1
PING 127.0.0.1 (127.0.0.1) 56(84) bytes of data.
64 bytes from 127.0.0.1: icmp_seq=1 ttl=64 time=1.61 ms
64 bytes from 127.0.0.1: icmp_seq=2 ttl=64 time=0.045 ms
64 bytes from 127.0.0.1: icmp_seq=3 ttl=64 time=0.100 ms

--- 127.0.0.1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2019ms
rtt min/avg/max/mdev = 0.045/0.584/1.607/0.723 ms

┌──(osboxes㉿osboxes)-[~]
└─$ 
```